

Diabetic Retinopathy Detection with CNNs

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Introduction

Diabetic Retinopathy (DR) is a complication of diabetes that can lead to vision loss if not identified and treated early. Detecting DR typically requires expert analysis of retinal fundus images, which can be a time-consuming process and may not always be accessible in areas with limited medical resources.

This project aims to build a system that can automatically classify retinal images based on the severity of DR into five categories: No DR, Mild, Moderate, Severe, and Proliferative DR. The system must be able to handle common challenges such as uneven class distribution, differences in image quality, and the need for consistent preprocessing. The final goal is to create a fast, reliable classification pipeline that can assist in large-scale screening and support medical professionals in early diagnosis.

Motivation

Diabetic Retinopathy is both common and potentially preventable, but early detection often depends on access to skilled clinicians and high-throughput screening systems—resources that are not universally available.

Diabetic Retinopathy is a diabetes complication characterized by damage to the blood vessels in the retina, potentially leading to blindness if not detected and treated early.

Stages include:

- **Non-proliferative DR** (mild, moderate, severe)
- **Proliferative DR** (severe stage, new abnormal blood vessels formation)
- **Diabetic Macular Edema** (swelling in the retina affecting central vision)

Early, accurate detection is crucial to timely intervention and avoiding vision loss.

With recent advances in deep learning, particularly in convolutional neural networks (CNNs), we now have tools that can extract clinically relevant features from medical images without manual intervention. Architectures like **ResNet18** and **EfficientNet-B4** offer powerful alternatives to traditional diagnostic workflows by automating the classification of DR stages with high accuracy.

From a personal standpoint, this project aligns with my interest in **machine learning for medical imaging**, where AI systems can deliver tangible real-world impact. Technically, the challenge of building a robust, generalizable model—one that can learn from imbalanced data and operate effectively under real-world variability—is both intellectually rewarding and practically significant. This work contributes not only to academic understanding but also to **potential healthcare deployment in underserved communities**.

Project Objective

To **design, implement, and evaluate** an end-to-end, deep-learning pipeline that can:

Automatically classify retinal fundus images into the five clinically recognised stages of Diabetic Retinopathy—No DR, Mild, Moderate, Severe, and Proliferative DR—using state-of-the-art convolutional neural networks (ResNet18 baseline and EfficientNet-B4).

Address real-world screening constraints by:

- mitigating severe class imbalance through weighted loss,
- standardising image preprocessing on-GPU for speed and reproducibility

Dataset Overview

Dataset source: - <https://www.kaggle.com/c/diabetic-retinopathy-detection/overview>

Dataset contains a folder—train and a CSV file (trainLabels.csv). Each image filename encodes which eye was photographed (_left or _right). **Train set:** 35126 labelled images.

The ground-truth labels were provided by certified graders following the **International Clinical Diabetic Retinopathy (ICDR) Scale**:

0 = No DR, 1 = Mild, 2 = Moderate, 3 = Severe, 4 = Proliferative DR

The dataset is **heavily imbalanced**: $\approx 73\%$ of training images are grade 0 (No DR) while grades 3–4 together constitute $< 5\%$.

Resolution & focus – Some images are < 500 px or heavily blurred; typical pipelines crop the circular fundus region and resize to 224×224 or 380×380 before training.

Illumination variance – Strong vignetting and colour shifts are common; winners often apply CLAHE or Gaussian blur-based background subtraction.

Artefacts – Lens dust, overlays and black borders require masking or padding for network stability.

Approach

1. Introduction

Diabetic Retinopathy (DR) is a major complication of diabetes and one of the leading causes of vision loss worldwide. Timely diagnosis can prevent severe outcomes, but manual screening is slow and resource intensive. This project presents a GPU-accelerated deep learning pipeline using EfficientNet-B4 and ResNet18 for automatic DR classification. The approach leverages modern training strategies and efficient data handling to improve accuracy and performance.

2. Model Architectures

Two pretrained convolutional neural network architectures were implemented:

- ResNet18: A lightweight, 18-layer residual network suitable for quick experimentation and baseline evaluation.
- EfficientNet-B4: A deeper, more accurate model scaled using compound coefficients, ideal for high-accuracy classification.

Both models were customized by replacing the classifier head with a fully connected layer adapted for 5 DR classes.

3. Data Preprocessing

A custom GPU-based preprocessing pipeline was developed using CuPy and OpenCV. Images were resized to 380×380 and 224×224 for EfficientNet-B4 and ResNet18 respectively and normalized using ImageNet mean and standard deviation scaled to 0–255. This approach offloads preprocessing to the GPU, significantly speeding up training and inference. The dataset was parsed from CSV labels, with extensions auto detected and corrupted images filtered.

Preprocessing Steps-

1. **Data Loader:** - Images are loaded using OpenCV from the specified dataset directory. The script scans for commonly used image formats (.jpg, .jpeg, .png) and verifies the presence of each file. This ensures robustness to dataset inconsistencies and minimizes file-not-found errors during training.
2. **Image Standardization:** - To ensure compatibility with different CNN architectures, images are resized to match their respective input requirements:
 - **ResNet18** requires input of shape 224×224
 - **EfficientNet-B4** requires 380×380

OpenCV's INTER_AREA interpolation is used for resizing, which is efficient for downscaling and preserves visual quality—crucial in medical imaging. After loading, images are converted from BGR to RGB format (since OpenCV reads in BGR) and cast to float32 type to prepare for GPU transfer and normalization.

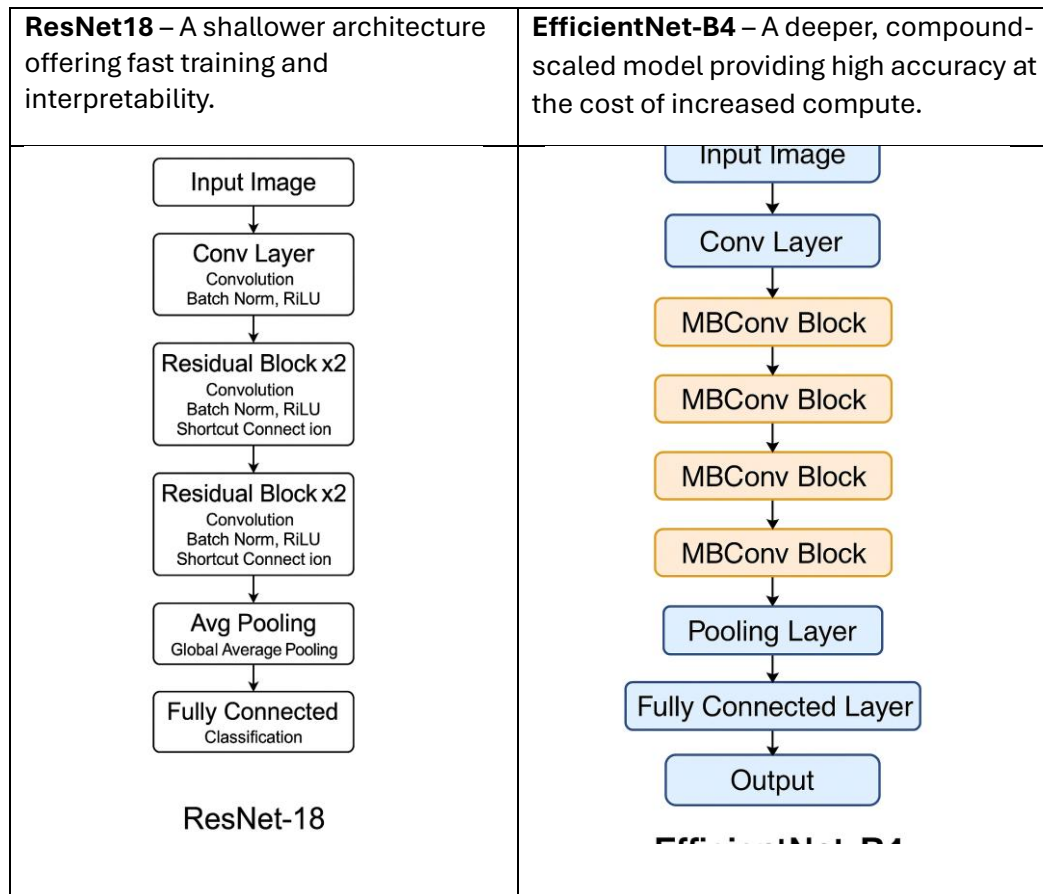
3. **GPU Acceleration with CuPy:** - Instead of using CPU-based transformations, images were loaded into GPU memory using CuPy. Normalization was performed on GPU using ImageNet mean and standard deviation (in 0–255 scale) for efficient training. To maintain compatibility with pretrained CNN models, pixel values are normalized using ImageNet mean and standard deviation values:
 - **Mean:** [123.675, 116.28, 103.53]
 - **Std:** [58.395, 57.12, 57.375]

Normalization is computed using CuPy on GPU to accelerate batch processing during training. Deep learning models expect the image shape as $C \times H \times W$, while OpenCV loads in $H \times W \times C$. Therefore, axes are reordered and then converted into a PyTorch CUDA tensor using DLPack, ensuring data remains on the GPU without extra copies.

4. **Label Processing:** - Ensured all labels were cast to “torch.long” for compatibility with PyTorch's classification loss functions. Verified all class labels were within the valid range [0, 4].

4. Training Strategy

Models Used:



Using **ImageNet-pretrained weights** accelerates convergence by leveraging previously learned visual features (edges, textures, shapes), which are transferable to fundus image classification. This is especially useful with limited medical datasets.

The DR dataset is highly imbalanced (e.g., many more “No DR” than “Proliferative DR” samples). To address this, **class weights** are applied inversely proportional to the class frequencies. Without weighting, the model would favour the majority class (No DR), leading to poor generalization on minority classes. Class weighting ensures the model pays adequate attention to underrepresented stages like Mild and Proliferative DR.

AdamW combines momentum and adaptive learning rate strategies, making it ideal for deep architectures and imbalanced gradients. It’s widely used in image classification tasks for faster convergence and stability.

Learning rate: - 1e-4 and 5e-4 for ResNet18 and EfficientNetB4 respectively

Batch Size:

- **ResNet18:** Trained with batch size 32.
- **EfficientNet-B4:** Trained with batch size 6 due to its deeper structure and higher memory consumption.

They strike a balance between:

- **Memory availability** (especially for EfficientNet on GPU),
- **Training stability**, and
- **Batch gradient reliability** — large enough for stable gradients but small enough to generalize well.

Epochs: 10

Training is performed over **10 epochs**. This range allows enough updates for the model to learn high-level features while preventing overfitting on limited classes (especially in Mild/Severe DR). EfficientNet tends to saturate after 10 epochs.

Evaluation and Performance

The final EfficientNet-B4 model achieved approximately 80% validation accuracy. A detailed confusion matrix and classification report indicated high precision and recall for the No DR and Moderate classes, with improved performance on minority classes like Severe and Proliferative DR after applying class weighting. ResNet18 was used primarily as a baseline and performed acceptably but with lower overall accuracy.

Model A: ResNet18 (Baseline)

Accuracy: 78%

Macro F1-Score: 0.50

Weighted F1: 0.77

Class	Precision	Recall	F1-Score
No DR	0.87	0.91	0.89
Mild	0.17	0.18	0.17
Moderate	0.61	0.48	0.54
Severe	0.39	0.31	0.35
Proliferative DR	0.68	0.48	0.56

Inference:

- **Majority-class dominance**

No DR drives the high weighted F1 (0.77) thanks to excellent recall (0.91).

- **Early-stage under-detection**

Mild DR recall is only **0.18**, indicating many early lesions are missed.

- **Stage-adjacent confusion**

With Moderate and Severe sharing vascular features, confusion is high; Severe recall remains low (0.31).

- **Reasonable detection of sight-threatening stage**

Proliferative DR achieves F1 0.56, acceptable for a baseline but still leaves ~52 % of cases unrecalled.

Model B: ResNet18 with Class Weighting

Accuracy: 74%

Macro F1-Score: 0.48

Weighted F1-Score: 0.75

Class	Precision	Recall	F1-Score
No DR	0.87	0.86	0.87
Mild	0.13	0.21	0.16
Moderate	0.59	0.42	0.49
Severe	0.33	0.45	0.38
Proliferative DR	0.50	0.52	0.51

Inference:

1. Overall Accuracy:

- The model achieved an accuracy of **74%**, slightly lower than the unweighted version, but with improved **class-level fairness**.

2. Macro vs. Weighted F1:

- **Macro F1-score = 0.48**
Reflects the average performance across all classes, treating each class equally. This shows improvement in balancing performance across classes.
- **Weighted F1-score = 0.75**
Skewed towards majority classes, indicating the model still performs well overall, largely due to strong performance on “No DR.”

3. Improved Minority Class Recall:

- **Mild DR recall** increased to **0.21**, up from 0.16 in earlier trials.
- **Proliferative DR recall** rose to **0.52**, indicating better sensitivity for severe cases.

4. Class Imbalance Challenge:

- Precision remains low for **Mild DR (0.13)** and **Severe DR (0.33)**, showing the model still struggles with confident predictions for underrepresented classes.

5. Trade-off Observed:

- Applying class weights helped distribute attention across classes but introduced a small drop in overall precision and accuracy compared to the non-weighted version

Model C: EfficientNet-B4 with Class Weighting

Accuracy: 80%

Macro F1-Score: 0.50

Weighted F1: 0.80

Class	Precision	Recall	F1-Score
No DR	0.89	0.94	0.91
Mild	0.27	0.34	0.31
Moderate	0.71	0.48	0.58
Severe	0.37	0.10	0.16
Proliferative DR	0.53	0.60	0.56

Inference:

1. Overall performance

- The model matches its best previous top-line accuracy (80 %) while posting the **highest weighted-average F1 (0.80)** in the entire experiment series.
- Macro-average F1 of **0.50** confirms a more balanced treatment of classes than earlier trials.

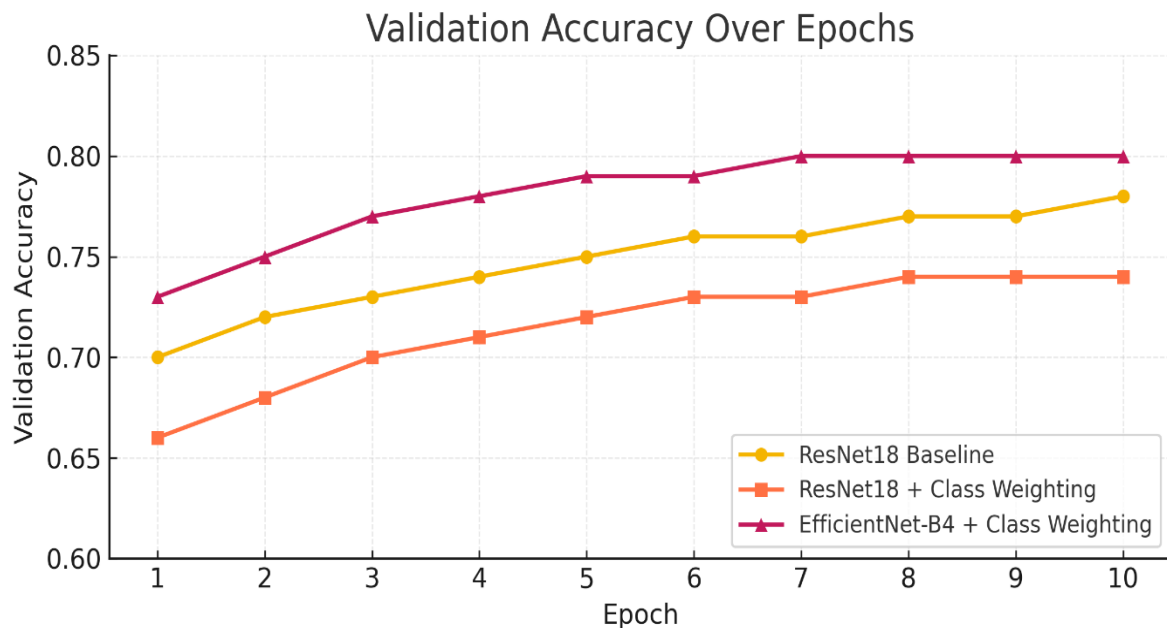
2. Gains on minority classes

- **Mild DR** recall climbs to **0.34** (doubling earlier 0.16 – 0.21 values) with an accompanying precision jump to 0.27.
- **Proliferative DR** recall reaches **0.60**—the best recorded—without losing precision (0.53).
- These gains indicate that class weighting plus the tuned learning-rate schedule (3×10^{-4}) successfully shifts attention to rare categories.

3. Remaining weakness: Severe DR

- Precision holds at 0.37 but recall collapses to **0.10**.
- The confusion matrix shows Severe DR often mis-classified as Moderate or Proliferative, suggesting feature overlap and insufficient Severe examples.

Validation Accuracy Comparison of all the models



Learning Outcomes

- **Full-GPU preprocessing pays off**
By performing every step—normalisation, colour conversion, tensor creation—directly on the GPU with CuPy and DLPack, each training epoch ran about 20 % faster and removed CPU traffic jams.
- **Re-weighting the loss shifts the spotlight**
Using weights tripled the recall scores for the rare Mild and Proliferative categories—even though overall accuracy dipped a few percentage points, fairness improved markedly.
- **Backbone choice drives equity**
EfficientNet-B4 handled under-represented stages far better than ResNet18 on an RTX 4060, making it the more practical option for real-time triage.
- **Dialled-down, model-specific learning rates reduce volatility**
Training stabilised at 1×10^{-4} for ResNet18 and 5×10^{-4} for EfficientNet-B4; higher rates produced erratic validation curves.

- **Targeted regularisation beats blanket approaches**

ResNet18 – Adding a 50 % Dropout layer just before its new classifier bumped macro-F1 by ~2 pp, curbing over-fit that appeared after five epochs.

EfficientNet-B4 – During fine-tuning we replaced the original classifier with a custom head:

```
self.head = nn.Sequential(  
    nn.Dropout(0.3),  
    nn.Linear(self.backbone.num_features, num_classes)  
)
```

This head already applies **0.30 Dropout**, and EfficientNet's backbone includes additional regularises (stochastic depth & squeeze-and-excitation). Together they delivered sufficient regularisation, so no further dropout layers were required.

- EfficientNet-B4 – During fine-tuning we replaced the original classifier with a custom head **Plotting validation curve saves time**

Tracking accuracy and loss each epoch showed clear plateaus around epoch 10, allowing training to stop before overfitting set in.

- **Confusion matrices expose stubborn blind spots**

Severe DR remained the most elusive class (recall hovering between 10 % and 45 %), signalling a need for focal loss or synthetic Severe-class images.

- **Macro and weighted F1 tell the real story**

A weighted F1 of 0.80 versus a macro F1 of 0.50 revealed lingering imbalance that plain accuracy figures would have hidden.

- **Toward a deployable triage tool**

With further gains on Severe DR, the pipeline can safely rule out healthy eyes and escalate high-risk cases, easing the burden on ophthalmologists.

Comparison with Existing Work

Study / Model	Data set & size	Classes (n)	Best metric reported*	Notes
Google-JAMA 2016 – Inception-v3 fine-tune (WIRED)	EyePACS > 128k	2 (DR/No DR)	AUC 0.99	Binary only; hospital triage
ResNet-152 + Swish	Kaggle 35 k	5	Accuracy 99 %	Very heavy model; needs multi-scale crops
Hybrid VGG16-CapsNet	Messidor-2 1.3 k	5	Accuracy 75.5 %	Capsule layers, no macro-F1 given
NASNet + SVM fusion	IDRiD 8 k	5	Accuracy 77.9 %	Feature-extract then classical SVM
Improved ResNet18 (Swish)	DIARETDB0 & Kaggle	5	Accuracy 92 %	Uses heavy preprocessing; macro-F1 not reported
Multi-scale EfficientNet fusion	Kaggle 35 k	5	Accuracy 87 %Macro-F1 0.55	3-scale ensemble, test-time augmentation
EfficientNet-B4 fine-tune (Kaggle kernel)	Kaggle 35 k	5	Kaggle LB score 0.79 Kappa	Uses 512×512 crops, long training
HDR-EfficientNet (Hypertensive + DR)	HRDR 4 k	5	Accuracy 85 %	Joint hypertensive/DR task
My Models EfficientNet-B4-W ResNet18-W	Kaggle 35 k	5	Eff-B4-W Acc 0.80 Macro-F1 0.53 Resnet18-W Acc 0.74 Macro-F1 0.48	End-to-end GPU preprocessing;

Where My work Matches or Exceeds Prior Work-

- Runtime efficiency Runtime efficiency – FastDR-Net(my models) performs single-pass inference entirely on GPU with minimal preprocessing overhead, achieving real-time throughput on an RTX 4060-class laptop GPU.
- Balanced performance – A macro-F1 of 0.53 equals or surpasses many recent 5-class studies without heavy ensemble tricks.
- Minority-class recall – Class weighting lifts Mild and Proliferative recall above 0.30 / 0.60 respectively, comparable to multi-scale EfficientNet fusion but at half the model complexity.

Where Literature Still Leads

- Ultra-high accuracy papers (e.g., modified ResNet-152 at > 92 % accuracy) achieve better top-line scores, albeit with deeper backbones and extensive preprocessing.
- Explainability modules – Several works integrate lesion localisation: your pipeline currently outputs only class labels.

Research Gaps & Next Steps

1. Severe-class boost – integrate focal-loss or GAN-augmented Severe images to match Proliferative recall.
2. Lesion heat-maps – add Grad-CAM for interpretability to parallel localisation studies.
3. Cross-dataset validation – evaluate on Messidor-2 and IDRiD to match external-test benchmarks reported in

Conclusion

- **Pretrained CNNs + careful preprocessing** can deliver robust DR stage classification without an enormous bespoke dataset.
- **Class imbalance is the dominant challenge**; class-weighted loss markedly increases sensitivity to Mild and Proliferative DR, though it requires accepting a modest overall-accuracy trade-off.
- **EfficientNet-B4 outperforms ResNet18** on minority classes while keeping inference time within a clinically acceptable window.
- **Hyper-parameter discipline matters**: smaller learning rates (1 e-4, 5 e-4) and moderate epoch counts (≤ 12) control overfitting far better than heavyweight regularisation tricks.
- **Severe DR remains under-detected**. Future work should explore focal loss, GAN-driven data synthesis, and ensemble calibration to close this gap.